

Hydrothermal breccias in vein-type ore deposits: A review of mechanisms, morphology and size distribution

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Abstract

Breccias are among the most widely distributed rock textures found in hydrothermal vein-type deposits. Previous studies have mainly been interested in developing qualitative descriptive approaches, leading to a confusing profusion of terms. Brecciation originates in numerous ways, resulting in highly complex classification systems and frequent misinterpretations of facies. Field observations are difficult to reconcile with physical theories of fragmentation, partly due to the fact that few satisfactory quantitative tools have been developed. A review of the main brecciation processes occurring in hydrothermal vein-type deposits allows for the discrimination between chemical and physical mechanisms, including tectonic comminution, wear abrasion, two types of fluid-assisted brecciation (hydraulic and critical), volume expansion or reduction, impact and collapse. Each of these mechanisms can be distinguished using nonscalar parameters that describe breccia geometry, including fragment morphology, size distribution of the fragments, fabric, and dilation ratio. The first two parameters are especially important because: (1) the morphology of the fragments allows chemical and physical (mechanical) breccias to be distinguished, and (2) the particle size distribution (PSD) is a function of the energy input during breccia formation. The slope of the cumulative PSD (fractal dimension) ranges from high values for high-energy brecciation processes, to low values for low energy processes indicated by an isometric distribution. The evolution of a vein system can be divided into three stages: propagation, wear and dilation. These stages are separated by one threshold of mechanical discontinuity and one of hydraulic continuity. These two thresholds also mark the transition between different types of brecciation. Mineralization occurs during all three stages and may display different textures due to pressure variations. The use of quantitative parameters in fault-related hydrothermal breccias allows a better understanding of the physical parameters related to a vein environment, including structural setting and crustal level, as well as fluid–rock interactions. Recognition of the different breccia types could also be important during the early stages of mineral exploration. © 1997 Elsevier Science B.V.

Keywords: breccia; hydrothermal vein-type deposits; morphology; size distribution

1. Introduction

Breccias are among the most common features in ore deposits. They are associated with numerous types of ores, either of endogene or supergene origin, and in both subsurface and submarine environments.

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The study of breccias has therefore been of major interest for ore deposit geologists who have endeavored to use breccia features to reconstruct the chronology and mechanisms of deposition, and to classify or model a deposit. Brecciation processes share strong ties with several other subdisciplines in the earth sciences including sedimentology, structural geology, seismology, rock mechanics and volcanology. Breccias are most common in the highest, most fluid-saturated part of the crust (defined as the schizosphere by Scholz, 1990), where brittle deformation without cataclastic or elasto-frictional processes is dominant (Sibson, 1986). Hydrothermal breccias constitute a subclass of the breccia family, in which brecciated rock interacts with hydrothermal (typically water-rich) solutions.

Geologists and geophysicists have employed two contrasting approaches during the last 20 years. Geologists, especially structural and ore deposit geologists, have tried to establish a general classification for brecciated rocks. The numerous attempts have ranged from purely descriptive to genetic and have used a wide variety of criteria. Sibson (1977, 1986) reviewed the structural brecciation processes in fault zones and proposed a useful classification for the definition of fault rocks using textural features (e.g., rounding), internal clast deformation, clast size distribution, and clast and matrix composition. A major distinction was made between random and foliated breccias, suggesting that the former is characteristic of the upper crust and the latter of deeper levels. Cohesive and incohesive breccias correspond to low and high pressure environments respectively, and are defined using the final appearance of the rocks. However, many natural breccia systems do not correspond to Sibson's classification. For instance, reactions in hydrothermal systems between host rock and fluid may lead to either dissolution or crystallization, possibly resulting in the development of cohesive breccias at low, rather than high, pressures (Schmid and Handy, 1991). Sillitoe (1985) uses a practical classification for breccias in plutonic and hydrothermal systems within magmatic-arc environments using the abundance and petrographic composition of the matrix or cement, the shape of the elements, and the overall organization of the brecciated units. Laznicka (1988) made a thorough review of breccia structures and associated rocks. He examined the

wide range of settings for breccias in different geological environments and proposed a fully genetic classification which distinguishes between gravity, dynamic (earthquake and explosion), low duration (hydraulic- and strain-controlled), volume change, and chemical processes during brecciation events. He also used a descriptive approach, using the Universal Rudrock Code independent of genetic interpretations. Laznicka's work is of great value because it offers an incomparable amount of data on breccias. More recently, mineralized breccias have been described from a system perspective, where three components (fragments, matrix and open space) are used to distinguish between fall-down, push-up and break-up breccias (Taylor and Pollard, 1993). Corbett and Leach (1995) grouped ore-related hydrothermal breccias into magmatic-hydrothermal, phreatomagmatic and phreatic types, according to increasing distal relationship to a porphyry source and increasing input of meteoritic waters. There is, however, still no consensus concerning a rigorous approach for breccia description and interpretation. Considering the huge amount of available data, it is surprising that brecciation processes remain so poorly understood. This may in part be explained by the complexity of the problem, and by the lack of well-established connections with other disciplines in which the same mechanisms have been studied in the laboratory (e.g., tribology and mineral processing; Stachowiak and Batchelor, 1993).

Geophysicists, on the other hand, have been pursuing a different approach. They have demonstrated quantitative relationships between the thickness, the length, and the offset of brittle fault zones, indicating that brecciation mechanisms should follow some statistical laws (Scholz, 1990). Fragmentation has been modelled using theoretical physical relationships between the surface or size of the fragments and the energy input (Von Rittinger, 1867; Epstein, 1947; Hartmann, 1969; Allègre et al., 1982; Brown et al., 1983; Cheng and Redner, 1988; Nagahama and Yoshii, 1993). Geologists can apply these theories to naturally faulted rocks in order to better understand breccias. For instance, Sammis and Osborne (1982) and Blenkinsop (1991) used quantitative analysis to show that it is possible to distinguish between shear and extension modes in breccias associated with splay faults of the San Andreas Fault system.

Due to the complexity of natural processes, however, it is still difficult to reconcile field observations with physical theory. The gap between the two approaches is partly due to the lack of quantitative methods for the characterization of breccias, although several parameters are potentially useful. In particular, the analysis of particle geometry in a given matrix has been the object of several textbooks, including that of Coster and Chermant (1989) and Russ (1995).

This paper is a review of the main mechanisms of brecciation in hydrothermal vein-type deposits, and emphasizes fragment geometry and distribution. Several parameters are needed for a complete geometric characterization, including fragment morphology, particle size distribution, fabric, and dilation ratios. Although these four parameters should be considered, I will focus on only two: the shape of the fragments and their size distribution (particle size distribution = PSD). Although natural breccias are usually complex, the approach will be limited to simple cases of monomictic breccias (e.g., breccias in homogeneous rocks) and to fragmentation processes that do not involve previously fractured and cemented particles. Analytical methods using image analysis and concepts derived from fractal geometry are given in the Appendix A and are detailed in a companion paper (Bérubé and Jébrak, submitted). Examples will be chosen from three ore deposit types that have been studied in detail both in the field and laboratory: precious metal epithermal (Kamilli and Ohmoto, 1977; Berger and Bethke, 1985), precious metal mesothermal (Colvine et al., 1988), and low temperature fluorite–barite vein-type deposits (Jébrak, 1984a,b; Von Gehlen, 1987). Comparisons will be made with breccias found in the epigenetic Au–U–Cu Olympic Dam deposit where many characteristics of the deposit's northwestern sector strongly resemble a vein-type deposit (Reeve et al., 1990; Lei et al., 1995).

2. Elementary mechanisms of brecciation

Only a few basic physical mechanisms exist for fragmenting a rock, and on a first order approximation, it is possible to distinguish physical from chemical brecciation. Chemical brecciation, or corrosive

wear, is caused by selective dissolution, whereas the main mechanism for physical brecciation occurs when the amount of stress exceeds the brittle resistance of the material (Griffith mode). However, numerous processes occur during fracture propagation in a hydrothermal system, such as subcritical crack growth, which allow cracks to propagate below the strength limit of the rock by combining mechanical and chemical processes. Brittle fracturing may appear in response to a variety of stress fields of different origins with very different scales and relative timing (Table 1). The sources for the stress can be divided into two categories (Bott and Kusnir, 1984): renewable, which persists despite continuing stress relaxation and corresponds to tectonic activity, and nonrenewable, which can be dissipated by relief of the initial strain (e.g., bending stresses and thermal stresses). The corresponding processes will be called incremental and instantaneous respectively.

Eight main mechanisms of brecciation can be defined (Fig. 1 and Table 1). Tectonic comminution, fluid-assisted brecciation and wear abrasion are the most common and are widely represented in vein-type ore deposits. Volume reduction, volume expansion, impact, collapse and corrosive wear are usually less abundant. It is important that these mechanisms are not treated as discrete processes, but rather as parts of a continuum in geological environments.

2.1. Fracture propagation

Hydrothermal breccias develop early during vein formation in response to the fracture propagation process. This process is well documented in structural geology studies (Scholz, 1990) and is one of the most common mechanisms of brecciation. Fracture propagation can be observed at all scales, from millimeter-sized fractures to kilometer-long faults which 'enclose' fragments of the hanging wall and footwall rocks. Numerous interactions occur between microfragments during small-scale fragmentation and these interactions are collectively referred to as wear abrasion. Fracture propagation and wear abrasion together are the two components of tectonic comminution. Brecciation associated with fracture propagation generally develops in association with persisting stress, and failure occurs when some critical stress is attained. However, in cases of long term

loading, natural fractures can also grow at stress intensities below critical stress (subcritical propagation; Atkinson, 1984). Tensile and compressive failure may occur, but because rocks are much weaker under tensional rather than compressional forces, tensional processes will generally dominate.

Breccias associated with fracture propagation have been encountered in all brittle fault zones (Table 1). Breccias caused by brittle tectonic comminution typically display fragments with angular morphology, especially in tensile fractures (Sammis and Biegel, 1986). Morphological measurements of fragment in the West Rouyn Fault show their angular, nearly Euclidian shape, expressed by a low value of D_s (Fig. 2 and Appendix A for methodology).

Fragment size is highly variable. There are no quantitative measurements of the particle size distribution (PSD) related to fracture propagation in natural systems, therefore we will use detailed field and laboratory studies on fracture propagation as well as quantitative modelling to address the geometric parameters of the breccia formed by fracture propagation. Fault development comprises two steps: firstly, an initiation of an array of en-echelon cracks and secondly, linkage of the cracks to form larger faults (Segall and Pollard, 1983; Granier, 1985). En-echelon cracks commonly display a general periodicity, caused by a random distribution of defects, a low speed fracture propagation, and near-equilibrium in the system (Granier, 1985; Renshaw and Pollard,

Table 1
Geological and physical processes of brecciation in hydrothermal vein-type deposits

Process	Stress	Origin	Geology	Other names	Examples	References
Tectonic comminution	renewable	uniform and non-uniform stresses (tensile or compressive)	comminution in brittle fault zones	fault breccia, break up	St. Salvy (Zn, France), J. Aouam (Pb, Ag, Morocco)	Cassard et al., 1994; Jébrak, 1984a
Fluid-assisted brecciation (hydraulic)	pulse	uniform stress (mainly tensile)	almost every type of deposit	crackle break up	El Hammam (F, Morocco), Dreislar (Ba, Germany), Creede (Ag, Au, Colorado)	Jébrak, 1984b
Fluid-assisted brecciation (critic)	pulse	uniform stress (tensile)	lode gold deposit	implosive, spalling, break up	Silidor (Au, Québec), Victoria (Au, W, Australia)	Carrier and Jébrak, 1994; Forde and Bell, 1994
Wear abrasion	renewable	uniform to non-uniform stress (compressive)	shear zone	milled, break up	Silidor (Au, Québec)	Colvine et al., 1988; Carrier and Jébrak, 1994
Volume reduction	non-renewable	uniform stress (tensile)	mud cracks, cooling of silica sinter	break up, desiccation, thermal contraction,	Cirotan (Au, Java), McLaughlin (Au, California)	Jébrak et al., 1996
Volume expansion	non-renewable	Herzian stress	porphyry copper, diatreme	milled, explosive, decompressive, push up	porphyry deposits (Cu, Mexico), Waiotapu (Au, NZ)	Clark, 1990; Hedenquist and Henley, 1985
Impact	non-renewable	Herzian stress	collapse breccias, erosive wear	push up, fall down	Maine (F, France), Silidor (Au, Québec), Les Farges (Pb, Ba, France)	Carrier and Jébrak, 1994; Jébrak, 1984b
Corrosive wear	pulse	disequilibrium	high fluid–rock interactions	milled, pseudo-breccias, break up	Olympic Dam (Cu, U, Au, Australia)	Reeve et al., 1990; Lei et al., 1995

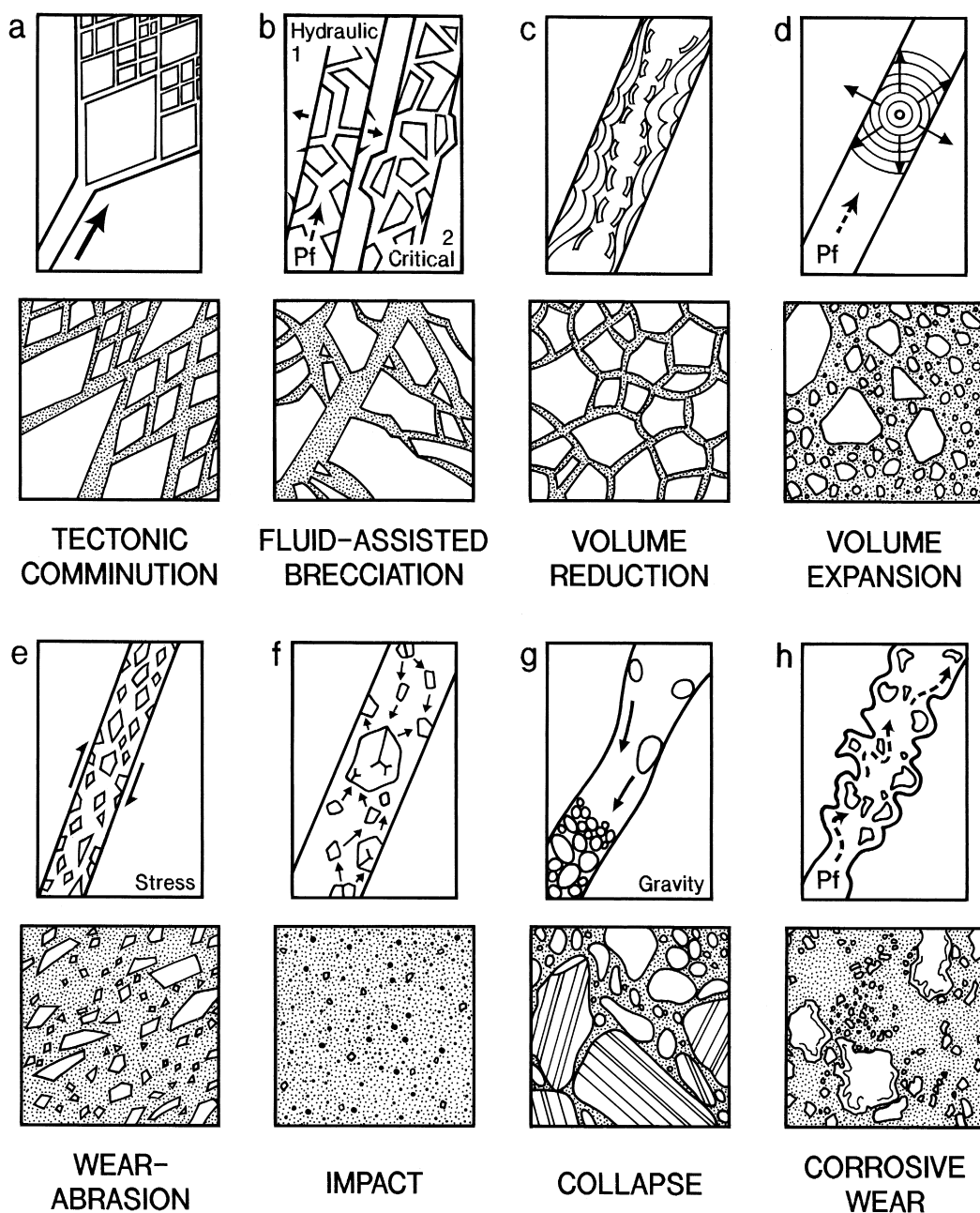


Fig. 1. Schematic illustration of the brecciation mechanisms in hydrothermal vein deposits, and resulting geometry of the breccias. Large arrow (tectonic comminution) indicates the direction of fault propagation. Small arrows indicate direction of displacement of the wall (fluid-assisted brecciation, volume expansion) or fragments (impact, collapse). P_f is fluid pressure. No scale is indicated, as most of the geometry is fractal.

1994; Wu and Pollard, 1995). This will lead to the formation of fragments of similar size and the PSD of large fragments within a fault zone will therefore

follow a normal law during the initiation step. As the system evolves, the propagation of the fracture may occur more rapidly because of the increasing fragility

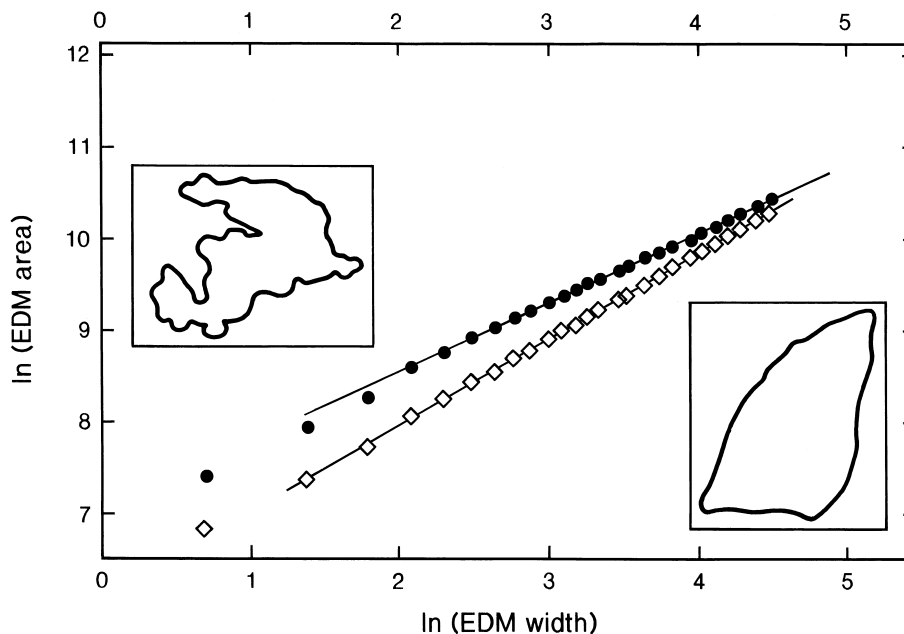


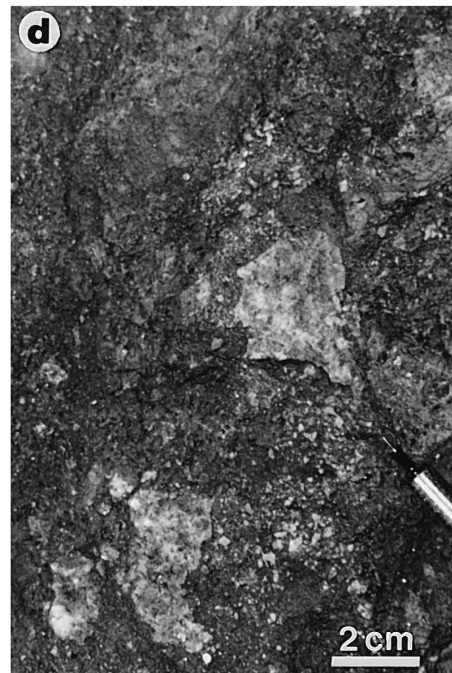
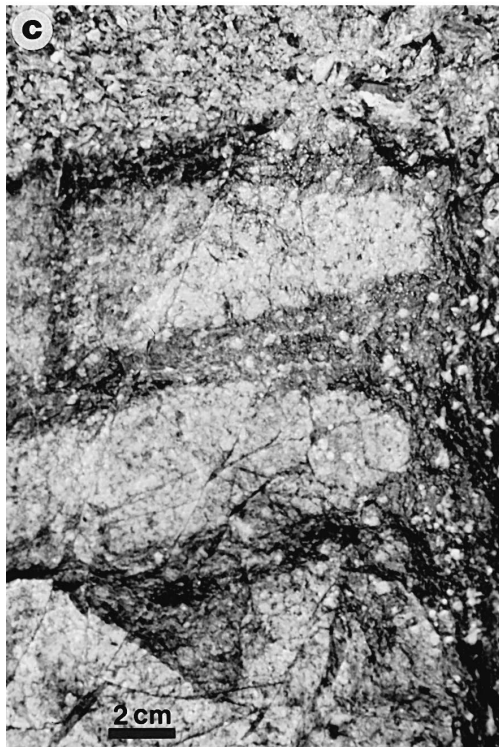
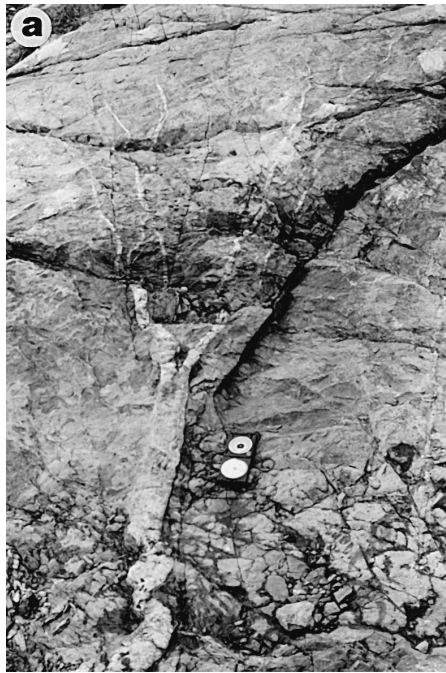
Fig. 2. The fractal dimension D_f of the morphology of particle is computed using the Euclidean distance mapping method (Russ, 1995; Bérubé and Jébrak, 1996). Ribbons of increasing thickness are computed from the particle outline. The log of the area of each ribbon is then plotted against the log of their thickness. Measurement of a hydraulic breccia from West Rouyn (Abitibi, Québec) (empty squares) and a chemical breccia from Olympic Dam (South Australia) (black dots).

of the media. Arborescent branching can occur during the propagation of the shear zone reflecting the instability of the propagation mechanism and the speed of propagation itself, related to stress intensity (Scholz, 1990). Arborescent processes have been observed in laboratory experiments and in the field, such as in Archean mesothermal gold deposits (Duvernay, Fig. 3a). The distribution of the spacing distance between fractures will follow a log-normal, exponential-negative or power law that has been frequently observed in hydrothermal systems (Huang and Angelier, 1989; Brooks et al., 1996; Johnson and

McCaffrey, 1996). Fragments formed by this propagation process will display a large variation in size that could result in a high value for the fractal distribution coefficient in a define range.

A directional fabric may appear due to the reorientation of the fragments parallel to the sense of movement, or at higher pressures, perpendicular to the main compressive stress axes. Dilation, defined as the ratio between matrix and fragment volumes, is typically low. Because tectonic activity typically has a longer duration than hydrothermal circulation, the conditions for mineral deposition may vary and sev-

Fig. 3. Photographs illustrating different morphology parameters in breccias of different origins: (a) Two types of brecciation in the Duvernay Au-deposit, Abitibi greenstone belt, Québec; below compass: hydraulic breccia in shear zone, with fragments of host-rocks set in a chloritic cement; note the regularity in size of the fragments, their angular morphology, and the small amount of displacement of the quartz vein in a plastic regime; left of compass, arborescent propagation of a quartz vein outward from the shear zone; (b) Collapse breccia in the Jebel Aouam Pb–Zn–Ag deposit, Hercynian Massif Central, Morocco; note the regularity in size and the rounding of ore fragments within the ankeritic cement; (c) Corrosive wear (diffusion-limited regime) in the Don Rouyn Cu–Au deposit, Abitibi greenstone belt, Québec; note the rounding of the dioritic fragments in a chloritized matrix; (d) Corrosive wear (kinetic regime) in the Olympic Dam U–Cu–Au deposit, Stuart Shelf, South Australia; note the angularity of the granitic fragments due to selective dissolution of the feldspars (hematized granite matrix).



eral mineral assemblages may characterize the breccia. These assemblages can be used to reconstruct the history of brecciation within the fault.

2.2. Fluid-assisted brecciation

Fluid is abundant at every level of the crust, especially in its more brittle part (Fyfe et al., 1978). In hydrothermal systems, the most frequent brecciation process is hydrofracturing, which is related to temporal variations in fluid pressure (Phillips, 1972; Jébrak, 1992; Hagemann et al., 1992). The process of fracture formation and disjunction of the fragments can be divided into two steps: hydraulic fracturing and critical fracturing (Fig. 1).

2.2.1. Hydraulic fracturing

Hydraulic fracturing is related to an increase in the fluid pressure within the vein (Phillips, 1972). This causes a decrease in the effective pressure, which can lead to fracture propagation. Most hydraulic fracturing is produced in an extensional regime, although it can also occur in a contractional environment (Beach, 1980). The increase in fluid pressure may have several origins, including a decrease in fault permeability due to fault slip or mineral deposition, and effervescence or boiling as a result of chemical reactions (Parry and Bruhn, 1990). Most of these processes are transient and brecciation will commonly mark one or several specific moments during mineral deposition. Brecciation can appear before or during vein formation, but because it will tend to preferentially occur in rocks with low permeability, it will frequently be observed at the beginning of the infilling process prior to extensive fragmentation (e.g., epithermal and low-temperature vein-type deposits; Table 1).

2.2.2. Critical fracturing

Critical fracturing is related to the destruction of the equilibrium between the pressure of the fluid and the regional stress within a vein (Hobbs, 1985). Fluid pressure decreases in response to a sudden opening of space generated by rapid slip or by the intersection between different veins. Any increase in the porosity of the system, especially after hydraulic fracturing, will provoke decompression and spalling instabilities on the vein wall. The best-documented

examples are implosion breccias in dilational jogs (gaps) formed by two opposing shears (Sibson, 1986; Forde and Bell, 1994), and at the intersection between two growing faults. This explains why cross-cutting veins are commonly associated with zones of intense brecciation. This type of brecciation typically develops during vein formation, and has been frequently observed in mesothermal gold deposits (Table 1).

Hydraulic and critical brecciation are always strongly associated because they are both associated with variations in fluid pressure. Both of these types of fluid-assisted brecciation generate in situ fragmentation textures (mosaic breccias) in a jigsaw puzzle pattern without significant rotation of the fragments, although rotation can often be observed in critical brecciation because the fragments generally collapse immediately following the fragmentation. This latter process may even appear in rather deep environments, such as mesothermal lode gold deposits where tilted host rock blocks demonstrate the initiation of collapse processes (Jébrak, 1992). An absence of rotation indicates that critical brecciation did not occur extensively, and that the dilation process in the vein was a transient phenomenon with a limited amount of open space.

In fluid-assisted brecciation, fragments are angular and brecciation typically follows pre-existing planes of discontinuity, like bedding or schistosity. This is related to the relatively low amount of energy required for hydrofracturing. Pressure fluctuation may also cause hypogene exfoliation which could locally lead to rounded fragments (Sillitoe, 1985). However, this type of process is usually a combination of both chemical and physical processes.

Although there is no systematic study of the PSD for fluid-assisted breccias, fragments are commonly of similar size (Jébrak, 1992; Fig. 4) with a lower fractal distribution coefficient (Blenkinsop, 1991). This may be related to the fact that the low level of energy required for this type of brecciation allows fractures with regular spacing to be developed (Renshaw and Pollard, 1994). The particles display much less overall comminution than particles in shear layers because the high fluid pressure in fluid-assisted brecciation preclude significant wear (Marone and Scholtz, 1989), and also because they typically form in an extensional regime. The extensional context is

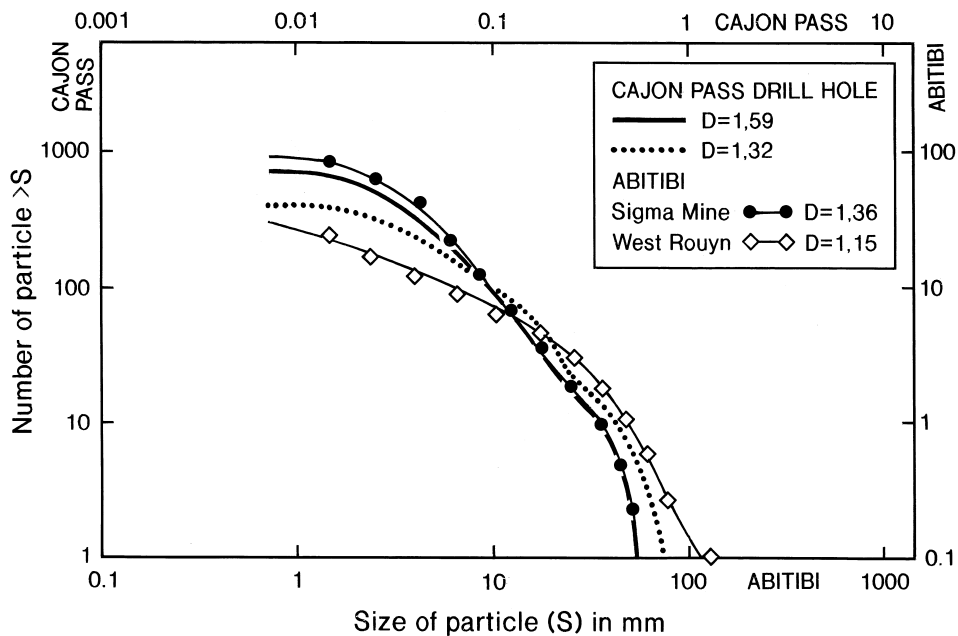


Fig. 4. The fractal dimension D_s of the particle size distribution is computed from the slope of the curve on an inverse cumulative histogram in a log–log diagram (see Blenkinsop, 1991 for more details). Measurements from the West Rouyn hydraulic breccia and the Sigma comminution breccia. High values correspond to very energetic processes of fragmentation.

also expressed by the abundance of matrix and typically high dilation ratios. The infilling material is usually simple, composed of only a few minerals, and banding is lacking since the pressure fluctuations are marked by rapid mineral precipitation.

2.3. Wear abrasion

Wear abrasion, or friction, occurs whenever a solid object is loaded against particles of a material that have equal or greater hardness (Fig. 1). In vein-type deposits, it occurs following the propagation process. Quartz typically acts as a wearing agent because of its hardness. Several micro-mechanisms occur concurrently during wear abrasion, including small scale fractures, cutting and fatigue by repeated plucking. Grain plucking is strongly dependent on the grain size of the brecciated rock and can be very important for coarse-grained rocks (Stachowiak and Batchelor, 1993). Wear abrasion may evolve into cataclastic flow— a deformation mechanism that involves uniformly distributed microcracking combined with rotation and frictional sliding of the frag-

ments (Higgins, 1971; Paterson, 1978; Arthaud et al., 1996). The physics of these processes is very complex and far from fully understood, especially because of the different behavior of the system at different scales.

At the macroscopic scale, fabrics formed during abrasion can be confused with those of a ductile process, yet at the microscopic scale these fabrics are obviously produced by the rearrangement of an aggregate of rigid grains. It is possible, however, that the transition between brittle and ductile behavior may arise due to the microplasticity of a material in a high pressure and high fluid-rock ratio context (Scholz, 1990; Hewton, 1991). Such a process is common to all brittle geological environments, but since frictional strength increases with effective pressure (effective normal load; Byerlee, 1978), the process is more commonly developed at depth. Repetitive sliding could cause fatigue-related cracks to form, producing large wear fragments.

Fragments in wear-abrasion breccias will commonly display evidence of rounding by rotation in the media. Dissolution–recrystallization occurs un-

der pressure and will add textural complexity to the boundaries of the fragments. Wear abrasion is one of the few processes that produce a large variety of particle size distribution types, from normal to fractal. Natural fault gouge particles typically obey a power-law PSD, especially in homogeneous rocks such as granite (Engelder, 1974; Anderson et al., 1980; Sammis and Osborne, 1982; Sammis et al., 1986). This simple law generally explains 70 to 80% of the distribution and is related to the uniform probability of particle fracture, independent of their size or strength (Epstein, 1947; Sammis et al., 1986; Sammis and Biegel, 1986; Sammis et al., 1987). Two fractal limits arise in gouge: a lower limit around 10 μm due to the dominance of intracrystalline porosity and mineral cleavage, and an upper fractal limit in the order of one centimeter (Sammis et al., 1987). The fractal dimension (D_s , see Appendix A) values are around 2.6 in gouge with submillimeter particles. Higher D_s values are observed if there is a selective fracturing of larger particles (Blenkinsop, 1991). D_s increases with the number of fracturing events, energy input, strain and confining pressure (Turcotte, 1986; Marone and Scholtz, 1989). An increase in the confining pressure does not always alter D_s , but does modify the value of the fractal limits and the mean grain size for a given scale of observation. Theoretical models based on energy release predict a power-law PSD with a D_s between 2 and 3, using either the relationship between fragmentation energy and the creation of new surfaces (Von Rittinger, 1867), or the relationship between fragmentation energy and the reduction in fragment size (Kick, 1885). However, none of these models are fully satisfactory because of their oversimplification (Nagahama, 1991).

In mineral processing, wear abrasion is a commonly used process and allows a large variation of the particle size distribution to be obtained during grinding, and even bimodal distribution has been observed (Harris, 1966). In natural systems, Engelder (1974) noted that some gouge does not follow a complete power-law distribution due to the natural limit attained when fragment size is equal to that of the rock pores. During wear abrasion some fragments maintain their initial size while others are reduced. This will destroy the uniformity of the PSD, forming several domains in the slope of the cumula-

tive curve, and it will not be possible to compute a unique fractal dimension. There are therefore very few specific PSD values for breccias related to wear abrasion. However, wear-abrasion breccias will usually display a distinctive fabric, which is related to the contrasting strengths of the different rock constituents. Brittle flattening (formed by the crushing of large particles) and subsequent redistribution will give the appearance of foliation. The amount of dilation is generally low or negligible.

2.4. Volume reduction

Fragmentation by volume reduction is not a common process in natural systems, but can occur as a result of phase transitions or temperature variations (Sibson, 1977). The most common volume reduction process is desiccation, which seldom occurs in hydrothermal systems. Evidence of desiccation processes has, however, been observed in some epithermal deposits (e.g. McLaughlin, California, and Cirotan, Indonesia; Table 1) owing to a periodic influx of silica supersaturated fluid followed by rapid drying (Laznicka, 1988; Jébrak et al., 1996). Desiccation is a form of brittle fracturing characterized by a polygonal network of extensive joints and is a transient process (Fig. 1). During the contraction of a layer of homogeneous material, desiccation creates a tessellation pattern and produces cracks perpendicular to the cooling or shrinking surface. It will produce identical brecciation to that formed by desquamation processes in the surficial environment (Bertouille et al., 1979).

Fragments of approximately the same size characterize the particle size distribution of desiccation breccias. Crack networks are not fractal because contraction-crack polygons generally have a characteristic length related to the elastic properties and thickness of the contracting medium (Korvin, 1989). Volume reduction brecciation can be modeled using joints located between a set of randomly distributed points (anticlustered Voronoi distribution; Budkevitch and Robin, 1994). The difference between the number of small and large fragments will be low, resulting in an almost horizontal slope of the curve on a log-log diagram (Appendix A). The low value of the parameter (D_s close to 1) will indicate a nearly Poissonian distribution.

Fragments display angular morphologies with 4 to 6 sides, and the fracture network will commonly be superimposed on pre-existing joints. Angles between fragments are usually about 120° . Dilation is generally of limited development and without multistage infilling.

2.5. Volume expansion

Volume expansion is generally related to transient explosion phenomena (Fig. 1). This mechanism is related to an unusual stress field, called a Hertzian stress field, where σ_1 (maximum principal stress) decreases progressively from the center of propagation of the fracture. This type of stress field can be generated by an explosion, or by the impact between an indenter and a surface (Frank and Lawn, 1967). The crack growth is orthogonal to the most tensile principal stress (σ_3) and corresponds to a surface delineated by the trajectories of σ_1 and σ_2 . Cracks may deviate from the stress path. In exceptional cases, Hertzian fractures can develop as the result of thermal stress (Bahat, 1977). Explosions can be provoked by chemical reaction, rapid decompression (Clark, 1990) or phreatic explosion, although true explosion-related breccias are rarely found in hydrothermal veins.

Volume expansion breccias are characterized by curved joints such as those in porphyry copper deposits (Clark, 1990; Table 1). Experimental work shows that rocks choked by a transient explosion process (like blasting or an atomic explosion) display a fractal distribution of fragment sizes with a very high ratio of large to small particles that is expressed by a high D_s value between 4 and 6 (Grady and Kipp, 1987). The large number of small fragments relative to big fragments is characteristic of abundant powder formation. No preferred orientation is generally observed, but a slight reorientation of large fragments may arise as a result of parallelism of fractures far from the center of the explosion. Dilation is usually significant, but is dependent on the intensity of the explosion and the strength of the rock.

2.6. Impact brecciation and collapse

When the walls of a vein are far enough apart, particles removed from the wall by brecciation may

travel downward or upward in the fluid and be abraded. In this manner, a vein may act as an autogenous mill when dilation is large enough to allow some mobility of the fragments. The dominant physical process in such an environment involves particle–particle and particle–wallrock impacts. Cupelling and chipping will occur. Impact brecciation, also known as erosive wear, relates to the ballistic behavior of such fragments in the fluid, where multiple reflections of collisional shock waves cause brittle fracturing to develop within a Hertzian stress field specific to each particle. Each particle can record a complex and dynamic evolution, and all the kinetic energy of the impacting particle is converted into elastic energy. Such a ballistic fracture system is the most effective way of creating a finely fractured medium (Kelly and Spottswood, 1982).

In a vein-type system, the more important factors will be particle strength, size and impact velocity. Impact brecciation is highly effective for ductile materials because even a high speed fluid without particles can be erosive, as demonstrated by the damage done to airplanes while flying through clouds (Stachowiak and Batchelor, 1993).

Impact brecciation and wear abrasion can both occur during the same brecciation event ('attrition' of Harris, 1966). The geological distinction between these two processes can be made on the basis of fragment mobility and fragment brittle–ductile deformation. Impact brecciation has seldom been recognized as a major mechanism for breccia formation in hydrothermal veins, although it may be much more prevalent than previously realized. Hydrothermal media are usually very dynamic, especially near the surface (Hedenquist and Henley, 1985), and hydrothermal minerals commonly contain solid microinclusions which may be interpreted to be xenoliths formed as the result of erosive wear. In volcanic environments, the 'mill rock' commonly associated with massive sulfide deposits is composed of rock flour probably formed by impact brecciation (Franklin et al., 1981). Upward milling has also been observed in subvolcanic breccia pipes, such as Kidston (Queensland). However, these rocks remain fairly uncommon and impact breccias are typically restricted to transient events during which exceptional acceleration of the flow allows particles to migrate rapidly and usually upward.

Impact breccias may also be generated within hydrothermal veins during collapse processes which cause fragments to be transported downward (Fig. 1). Collapse breccias are usually the result of increased spalling of the vein walls. Fragments of the host rocks and early minerals may fall into the conduit and subsequently interact together. Examples are found in the Pb–Ag Jebel Aouam deposit (Morocco) where the mechanism of opening by transtension created large voids now filled by fragments of earlier minerals (Fig. 3b; Jébrak, 1984a). In the fluorite deposit of Maine (France), some collapse fragments from the paleosurface traveled 200 m downward (Table 1).

Impact and collapse breccias will display some similarities, such as their rounded morphologies and smoothness. The distribution of fragment sizes will differ depending on the origin of the particles. A normal PSD is commonly observed because transportation will be a function of the hydrodynamic diameter and will tend to sort the fragments by size (Wohletz et al., 1989). PSD analyses carried out on amorphous materials display a fractal dimension around 1 (Kaye, 1993). In the Cirotan deposit (Indonesia), PSD values are relatively low. Also in this deposit, graded bedding has been observed and is interpreted to have formed during a collapse rolling process (Genna et al., 1996). The tendency of fragments to affix themselves parallel to vein walls will produce an overall orientation of fragments, and collapse breccias could show imbricated fragments at the bottom of hydrothermal cavities. The amount of dilation will generally be large, but breccias remain typically fragment-supported.

2.7. Corrosive wear (chemical brecciation)

Sawkins (1969) first proposed the concept of chemical brecciation, and his idea was mainly applied to porphyry systems where chemical reactions promote explosions. Chemical brecciation is very common in natural and artificial environments (Sahimi, 1992). The product is known as a solution breccia or a pseudobreccia (Jébrak, 1992; Fig. 3c). In tribology, the process is called corrosive wear and this term will be used here (Stachowiak and Batchelor, 1993). In the lithosphere, salient examples are produced by magmatic brecciation, hydrothermal

processes and the rounding of granitic rocks during weathering. Sahimi and Tsotsis (1988) proposed a general model for corrosive wear by studying the consumption and fragmentation of porous coal particles. Two reaction–consumption end members were defined as the kinetic and diffusion-limited regimes. In the *diffusion-limited regime*, only the most exposed part of the solid matrix is reached and consumed as a reactant; corners are therefore much more easily dissolved than a flat surface, and so the external surface of the fragments will become smooth and fragments may ultimately take a spherical morphology. Such a process will occur if there is a strong chemical disequilibrium between the rock and the fluid. In the *kinetic regime*, the alteration–dissolution rate is limited only by the chemical reaction rate. A uniform alteration will conserve the overall morphology of the fragments and indicates the presence of a fluid with relatively low chemical reactivity.

In hydrothermal veins, chemical processes are plentiful. Corrosive wear may occur at different times during the infilling events. Any crushing process that significantly increases the reactive surface can enhance corrosive wear. Corrosive wear will give several types of fragment morphology. Diffusion-limited regime processes produce smooth fragments, whereas kinetic regime processes enhance the contrasting compositions of the fragments and result in more complex final morphologies. Highly complex morphologies related to kinetic regime corrosive wear are exemplified in the rocks at Olympic Dam, South Australia where high roughness values ($D_r > 1.3$) have been measured (Figs. 2 and 3d; Tables 1 and 2). The particle-size distributions for chemical breccias have been experimentally determined for coal (Sahimi, 1992). The fractal dimension (D_s) in brecciated coal is usually low and similar to that resulting from tension crack formation (mode I fracturing). Very little data has been obtained for hydrothermal systems.

Brecciation-related reorientation of fragments generally does not occur during chemical processes. However, dissolution may be superimposed upon a previous anisotropy (like mineralogical banding or earlier fractures) that controls the infiltration of the solution. Dilation values are usually low, but will increase as the alteration progresses.

2.8. Conclusion

Hydrothermal vein-type deposits are the sites for numerous types of brecciation processes. Their recognition is not always straightforward. Fragment geometry can be used as a tool for recognizing their diverse origins. Two geometric parameters appear to be especially useful for this purpose:

(1) Fragment geometry, either simple or complex, can help distinguish between chemical brecciation of the kinetic type and the various kinds of mechanical brecciation. However, it is not possible to identify an origin based only on a rounded fragment shape, since that shape may be related to several different brecciation processes, including fluid-assisted (hypogene decompression), impact or chemical (diffusion-limited dissolution).

(2) Fragment distribution allows several different processes to be distinguished. A PSD value close to 1 is indicative of fragmentation related to volume diminution. A low PSD value is often observed in hydraulic breccia or in breccias where transportation processes provoke a size classification. In such cases, the distinction between process should be made using other methods, such as the nature of the fragments. Higher PSD values indicate large differences in fragments size that correspond to tectonic fragmentation (tectonic comminution or wear abrasion) and can be distinguished using the amount of displacement between the fragments. Very high PSD values are typical of fragmentation by explosion processes (Grady and Kipp, 1987).

These two geometric parameters can be used to construct a classification diagram for hydrothermal breccias. For example, roughness (D_r) and PSD (D_s) can define fields for the different breccia-forming environments which account for the main breccia types (Fig. 5). Boundaries in the diagram are approximate because there is not enough data from natural examples and because of the considerable overlap between breccia types. This type of diagram could also be constructed for other geometric parameters which express the complexity of the geometry of individual fragments in relation to the type of particle size distribution.

3. Breccia evolution

When a medium is in a critical state, even minor adjustments to the system can have major repercussions. During vein formation, the host rock undergoes profound transformations as it changes from a cohesive to a fractured medium and then to a percolating medium. Propagation, two-media and three-media stages can be distinguished, each of which represent specific brecciation processes (Fig. 6). Two major thresholds separate these stages: a mechanical discontinuity threshold when the media becomes a noncontinuous solid, and a hydraulic continuity threshold when the fluid forms a continuously connected phase throughout the fracture system. The three stages can repeat in a cyclic manner in hydrothermal systems, allowing for very complex brecciation patterns to arise.

Table 2
Main characteristics of breccias in hydrothermal veins

Stage	Type	Mechanism	Duration	D_r	D_s (PSD)	Fabric	Dilation
PW	tectonic comminution	increase in regional stress	persistent	low	< 2	common	very low
P	hydraulic	increase in fluid pressure	periodic	low	< 2	none	high
W	critic	decrease in fluid pressure	periodic	low	$< 2?$	inherited	very high
D	volume reduction	temperature decrease	transient	low	≈ 1	none	low
W	volume expansion	pressure decrease	transient	low to medium	> 3	scarce	very high
W	wear abrasion	friction	transient	low to medium	non fractal	common	low
D	erosion	particle impact	transient	low	< 2	none	?
D	collapse	gravity	transient	low	≈ 1	scarce	high
WD	corrosive wear	dissolution (kinetic or diffusion-limited)	persistent	high	< 2.5	inherited	variable

D_s is fractal dimension of the particle size distribution.

First column: P = propagation stage; W = wear stage; D = dilation stage. 'Low D_r ' corresponds to values less than 1.1; 'medium' between 1.1 and 1.2, and 'high' more than 1.2. The dilation ratio is equivalent to porosity at the time of fragmentation.

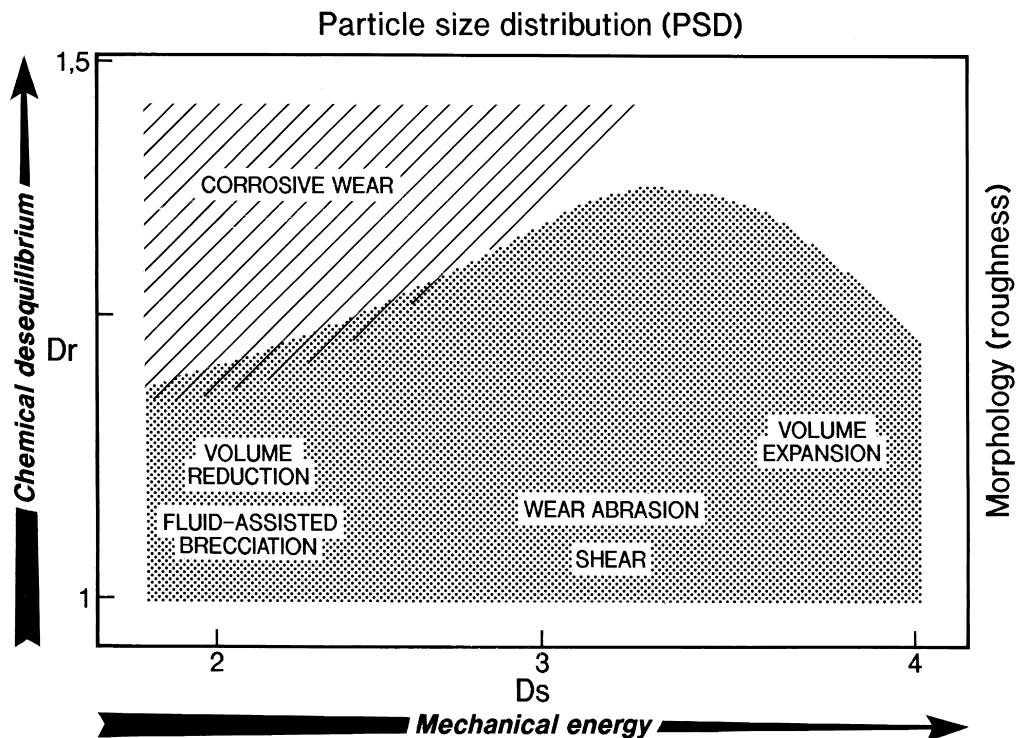


Fig. 5. Diagram of D_r (roughness fractal dimension) vs. D_s (particle size distribution) showing the approximate fields of the different types of breccias in hydrothermal vein-type deposits. Hatched zone (corrosive wear field) is based on limited measurements in the Olympic Dam and Don Rouyn deposits. Stippled zone (mechanical breccias) is based on measurements in the Cirotan deposit (Genna et al., 1996) and values from Grady and Kipp (1987).

The mechanical discontinuity threshold marks the transition from a solid medium to an assemblage of fragments. At this stage, there is no longer a continuous cohesion throughout the wall rock, and stress and strain within the medium become much more heterogeneous than before. The hydraulic continuity threshold marks the transition from an assemblage of discrete pockets of hydrothermal fluid separated by zones of interconnected wall rocks, to a permeable aquifer. During this stage, fluid pressure will be highly variable, and may even evolve from lithostatic to hydrostatic if the fault is connected to the surface. Thermal and chemical reequilibrations between individual fluid pockets may occur and hydrothermal solutions can be transported along the entire length of the fault.

3.1. Propagation stage

The propagation stage involves the nucleation and growth of fault patterns. The main mechanisms in-

involved are tectonic comminution, fluid-assisted brecciation and, less commonly, volume expansion by hydrothermal explosion (Fig. 6). Fracture propagation can be enhanced by stress corrosion, a mechanism that involves alteration at the tip of a fracture, thereby inducing corrosive wear.

Fracture initiation is a highly nonlinear process, and the entire process is strongly dependent on the pre-existing anisotropy or heterogeneity of the rock. Fracture propagation creates the abundant and widespread breccias associated with faults. In natural systems, brecciation is either caused by a rapid release of energy or a more long-term renewable process (Bott and Kusnir, 1984). If the energy input is generated by nonrenewable stress for a short period of time, brecciation could be a one-step process, as in the case for explosions and implosions. Volume reduction processes belong to the same one-step process because cooling and desiccation are almost instantaneous in a geological time frame. If the

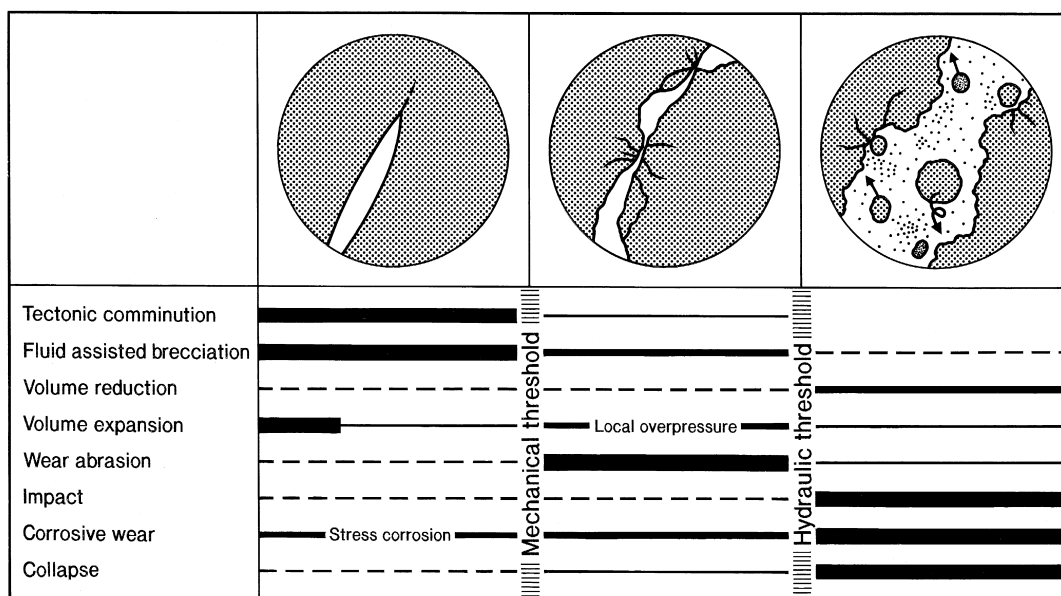


Fig. 6. Three stages of breccia formation within hydrothermal veins: propagation, wear, and dilation stages, separated by two thresholds related to mechanical discontinuity and hydraulic continuity. Thickness of the bars in lower chart express the relative abundance of the type of breccia during the three stages based on numerous examples from mesothermal gold deposits, epithermal Au–Ag deposits, and low temperature F–Ba–Pb–Zn deposits [see Table 1, and from literature, especially Laznicka (1988)].

energy input is protracted in response to renewable stress, incremental fragmentation will occur whereby the initial fracture event is followed by a series of other fracture events.

On another hand, brittle comminution may be associated with either quasi-uniform or nonuniform directed stress. Fig. 7 is a two-dimensional illustration of the four hypothetical different stress configurations that lead to different fracture patterns, either instantaneous or incremental. During quasi-uniform stress, σ_1 , σ_2 and σ_3 remain constant throughout the media. This may be accomplished in a purely tensile system associated with a regional stress, or in an extensional fracture network caused by volume reduction. Most of the strain during brittle comminution will be accommodated by numerous simple tension cracks typically accompanied by minor branching (Bahat, 1980). Local stress variations at the microscopic scale may form in response to fracture propagation, but such variations are generally minor. In such systems, strain in a perfectly homogeneous media can become localized with some periodicity due to the redistribution of the stress under subequilibrium conditions. Such periodicity is not

only predicted by mathematical modeling (Hafner, 1951; Couples, 1977), but is also observed during experiments (Wu and Pollard, 1995) and in ore deposits at various scales (see for instance Kutina et al., 1967; Valenta, 1989). Regular jointing will form fragments of approximately the same size, producing a Gaussian PSD. For nonuniform stress, σ_1 , σ_2 and σ_3 vary greatly in either their direction or their intensity due to external or internal causes. This could occur during explosion or shearing. Shear fractures are initiated by the formation of tensile cracks (Cox and Scholz, 1988) which then control the development of breccias. Both the rotation of the fragments and the rotation of the stress field during a noncoaxial deformation event will lead to a strongly nonuniform stress field at the shear zone scale; it is therefore suggested that the resulting pattern will be much more heterogeneous than that in an extension fracture pattern system during the propagation stage.

The intensity of the stress difference largely controls the velocity of crack propagation, and therefore exerts a controlling influence on fracture length, distribution and spacing (Renshaw and Pollard, 1994). The abundance and size of the fragments may

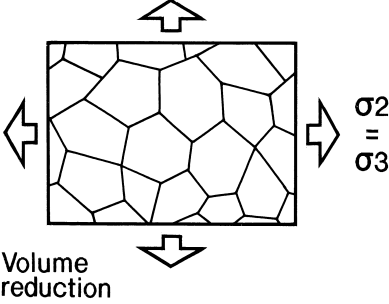
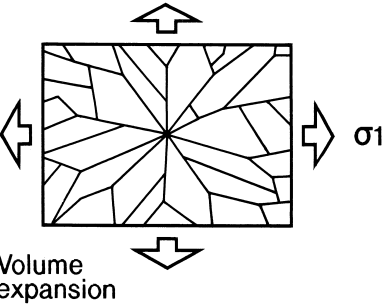
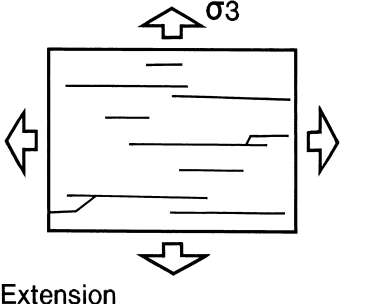
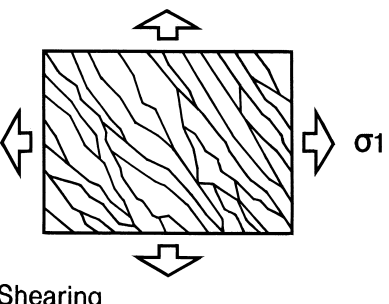
Stress Crack propagation	Uniform	Non uniform
Instantaneous		
Incremental		
Fragments distribution	Normal	Fractal

Fig. 7. The two parameters of stress and crack propagation play a major role during the propagation stage, and define early fragmentation in a vein-type deposit. Crack propagation can be instantaneous, related to transient, nonrenewable, stage of stress, or incremental (constant state of stress). The stress field could be relatively uniform, like in a tensional environment, where few rotations occur, or nonuniform, like in a compressional environment. Arrows indicate one of the principal stress directions.

be related by applying the Euler topological law to the length and spacing of their boundaries (i.e., the fractures) (Coster and Chermant, 1989). At low stress difference intensities, nucleation of fractures is a slow process. It allows for a redistribution of the tension within the rock because the formation of a fracture is associated with a decrease of the nearby stress field, which in turn reduces the probability that another fault will grow nearby. This feedback process leads to a self-organized fault periodical distribution. Such a model is similar to that of Orange et al. (1994) which explains the regular canyon spacing along passive margins. A periodicity in the fracturing process may therefore appear in a low intensity

uniform stress field. In such a context, fracture density follows a normal quasi-Gaussian distribution, and fragments will display a fairly homogeneous size distribution (Olson, 1993). Low propagation velocities appear during extensional fracturing, especially if a fluid is present (hydraulic breccias, Fig. 1), or during subcritical crack propagation (Atkinson, 1984). Such an extension could be bi- or tri-axial. If $\sigma_2 = \sigma_3$, the pattern will evolve toward a homogeneous joint system, as exemplified by the cooling pattern of basalt, glacial ice-wedge polygons, or mud cracks (Lachenburg, 1962; Stevens, 1974). If $\sigma_3 < \sigma_2$, the pattern will reflect the nucleation of subparallel joint sets.

As the energy of the brecciation process increases, the velocity of fracture propagation should also increase, and more interactions between fractures will occur (Renshaw and Pollard, 1994). The fracture distribution will become progressively skewed, with numerous small fractures and few large ones. This will constitute a population of fragments with a fractal PSD and high D_s values (Turcotte, 1986). This could occur in two cases: (1) an explosion process, and (2) a shearing event. An explosion process corresponds to tensile cracking over a very short time interval within a nonuniform Hertzian stress field (Frank and Lawn, 1967). For example, superheated water may initiate a very rapid phase of nucleation causing steam brecciation to occur. The breccia development will display all the characteristics of a chaotic process and is strongly dependent on the initial rupture: slight variations during the nucleation process or crack growth will have important consequences on the final distribution of the fragments. Shearing within the shear zone corresponds to a compressive cracking regime, and is generally caused by a nonuniform stress which varies in intensity and orientation during the propagation process (Chinnery, 1966).

Although it remains speculative, the propagation mode could therefore define, at an early stage in the fracturing process, the morphology and size distribution of the fragments. At low energy levels, and especially in the presence of fluid, redistribution of stress will allow for the formation of isometric breccias, whereas at higher energy levels, fractal distribution occurs and coincides with a cascading distribution of energy release (Stanley, 1971; Schertzer and Lovejoy, 1990). Transitions from low-energy to high-energy brecciation occur in space and time, but because these two endmembers give rise to different types of joint or fault networks and particle size distributions, it is necessary to distinguish between them.

3.2. Wear stage

The wear stage represents the longest period of breccia formation in a hydrothermal vein-type deposit. After the initial propagation stage, the mechanical threshold is crossed and the medium becomes discontinuous. Displacements along the fault wall

usually occur several times before reaching complete hydraulic continuity of the medium. Such tectonic movements are related to the seismic cycle (Sibson, 1986) and earthquake rupturing, and are associated with limited fluid circulation and sealing by mineral deposition. Energy will be partitioned for both fracturing and the differential motion of the blocks, thus modifying the packing geometry and the porosity. Several mechanisms of brecciation will be operating, among which wear abrasion will be dominant (Fig. 6) although the large fluid pressure variations could also provoke fluid-assisted breccias. Reactions between the fluid and the host rock can cause local corrosive wear breccias to form, whereas sudden changes in the physical state of the fluid can create volume expansion (explosion) breccias.

Wear abrasion will occur because of the roughness of the vein walls and the evolving geometry of the conduit during this stage of breccia evolution. Also during this stage, an abrasion process resulting from the relative movement between the two sides of a fault will mostly control breccia formation. The abrasion process may be controlled either by the roughness of the surface of the walls or by the mineralogical and PSD characteristics of the gouge (Sammis and Biegel, 1989). Most of the PSD for the wear abrasion breccias associated with this stage should follow a fractal law (e.g., Sammis and Biegel, 1986, 1989; Marone and Scholtz, 1989) due to the formation of abundant small fragments and the eroded remnants of larger ones. However, as observed by Harris (1966) in experimental work and Engelder (1974) in natural systems, the PSD of wear abrasion related breccias can actually follow many different laws. Moreover, breccias generated during the wear stage typically reflect a pulsative process and any ore associated with this stage will commonly display multi-stage deposition and will generate a complex PSD, resulting from the cumulative effect of the numerous episodes.

The rapid changes in the geometry of the medium during the wear stage will promote fluid pressure fluctuations which can modify the effective stress and the physical state of the fluids. Fluid-assisted brecciation (hydrofracturing) will therefore be one of the common processes during wear stage and is indicative of the high pressures easily sustained by a medium with relatively low permeability. Fragments

may also collide with each other causing intraparticle fracturing and autogenous brecciation.

3.3. Dilation stage

The third stage of breccia evolution is characterized by dilation and appears after the second critical threshold when the fault network becomes connected and fluid percolation can occur. At this point, the medium becomes hydrodynamically continuous. Its permeability increases suddenly, with possible transitions from lithostatic to hydrostatic confining pressure. Several types of breccias may be formed during this stage, including those related to corrosive wear and open space physical processes (e.g., collapse and impact).

Corrosive wear is dependent on the time-integrated water–rock ratio. This ratio will rapidly increase after the establishment of hydraulic continuity, allowing hydrothermal fluids to percolate around each fragment. Impact brecciation and wall erosion may occur and will be facilitated by the availability of hydrothermal fluids that can chemically weaken the material. However, at the same time, the appearance of a vertical hydraulic gradient in the conduit allows the fluid to be transported with a higher flow rate and the residence time will be shorter. The results of such corrosive wear of the host rocks has been observed in low temperature fluorite–barite veins in the Hercynian chain, and especially in the Langenberg deposit (Jébrak, 1984b).

The most important processes will be spalling and collapse of fragments in the channelway (Fig. 6). Collapse breccias commonly observed in barite–fluorite or Pb–Zn–Ag vein-type deposits (Jébrak, 1992) and gold mesothermal deposits (Carrier and Jébrak, 1994) are mostly barren, which could reflect the dilution of mineralizing solutions by sterile fluids during a major change in the hydraulic flow. Others mechanisms of brecciation will be of lesser importance. Volume reduction is limited to very specific colloidal deposition and does not appear to play any major role.

The transition from a lithostatic to a hydrostatic regime induces instability along the wall rock and abundant collapse breccias may mark the later phases of hydrothermal vein-type deposits (Fig. 3b). The dilation stage could become pulsative if there is

strong cementation during the process (i.e., mineral precipitation) resulting in a cohesive breccia. Spalling from the vein walls is related to decompression events. The combination of accretion of hydrothermal minerals around fragments and collapse within newly formed cavities could lead to the formation of cockade breccias (Genna et al., 1996), with reverse graded bedding of the fragments.

3.4. Conclusion

The relationship between vein infilling and fault movement can be complex. In the simpler cases, without multiple stages, the deposition of economic minerals can be associated with any of the three stages of brecciation previously discussed. From the descriptions of the multiple mechanisms of breccia formation, the following generalizations are tentatively proposed:

(1) During the propagation stage, breccia formation is usually single stage and precipitation of the minerals is often related to a decrease in fluid pressure because the solubility of most ore minerals increases with pressure.

(2) Breccias generated during the wear stage typically reflect a pulsative process (Parry and Bruhn, 1990) and mineralization associated with this stage will commonly display multi-stage deposition.

(3) Pressure variations are of lower amplitude during the dilation stage because of the mechanical continuity of the medium. Mineral deposition during this stage will mainly be associated with independent pressure processes such as mixing or cooling.

4. Exploration implications

The detailed study of ore deposit textures and structures has always been an important and critical tool of economic geologists. Observations of deposit style combined with careful examination of vein textures can have significant implications for exploration (Vearncombe, 1993). Breccias can be helpful in deciphering key elements of ore deposits, including structural setting, crustal level and fluid–rock interactions.

Breccia textures can potentially be used to determine the crustal level at which they formed because

the rheology of rocks changes with temperature and pressure. Colvine et al. (1988) and Groves et al. (1991) have proposed models for mesothermal lode gold deposits in which breccias mark the upper part of the deposits. In these deposits, several kinds of breccias have been encountered. The most common are the tectonic, hydraulic and critical types related to fluid pressure variations likely caused by seismic pumping (Sibson, 1977). In more surficial precious metal epithermal-type deposits, hydraulic and collapse breccias are more abundant, the latter typically occurring in meter-scale openings (Genna et al., 1996). The fluorite–barite deposits that form near the paleosurface in uplifted crystalline basements display similar breccia types, although the ore deposition is related to a more surficial and less convective hydraulic system (Jébrak, 1984b; Von Gehlen, 1987). Desiccation breccias seem limited to the uppermost part of epithermal veins (Jébrak et al., 1996). Fluid–rock interaction is also expressed by the geometry and the mechanism of breccia formation. Pressure variations are indicated by the presence of hydraulic breccias. Corrosive wear is related to the degree of equilibrium between the composition of both the fluid and the rocks. The duration of the process also plays a leading role and it is necessary to use a reaction progress parameter to get a more quantitative evaluation of the fluid–rock interactions (Ferry, 1986).

The processes described in this paper are similar to those at work in explosive volcanic environments (Vincent, 1994). For example, breccia pipes correspond to a combination of explosion, collapse and shattering processes accompanied by igneous injection. The application of quantitative techniques should allow the relationship between volcanic and hydrothermal brecciation processes to be clarified. In mineral exploration, quantitative criteria for breccia recognition can be used at different stages:

- (1) The recognition of the type of brecciation process could lead to a better discrimination between deposit models. Numerous examples of this approach are given by Laznicka (1988) and Taylor and Pollard (1993);

- (2) Mapping quantitative geometric breccia parameters could be used as a new tool in exploration. For example, the transition from hydraulic to collapse breccia in fluorite–barite deposits coincides

with the root of the ore shoot (Jébrak, 1984b). This could be expressed by measuring the morphology of fragments in the veins; and

- (3) Relationships between brecciation processes and mineral deposition may provide detailed information about the stages of ore deposit formation, possibly allowing reconstruction of the paleopermeability of the system during mineral deposition.

5. Conclusions

Eight major mechanisms of physical brecciation can be distinguished in hydrothermal vein-type deposits: tectonic comminution, fluid-assisted brecciation (hydraulic and critical), wear abrasion, volume reduction, volume expansion, impact, and collapse. Corrosive wear corresponds to chemical brecciation. Each of these processes is characterized by a specific geometry, although transitions and overlaps do exist.

In order to fully describe breccia geometry, several parameters are needed, including fragment morphology, particle size distribution, fabric, and dilation ratios. The first two parameters have been used in this study. The morphology of the fragments allows chemical and physical (mechanical) breccias to be distinguished. The particle size distribution (PSD) is related to the energy input for breccia formation; high-energy brecciation processes tend to develop an anisometric PSD, whereas low energy brecciation processes typically develop an isometric distribution. The slope of the PSD (i.e. fractal distribution) is a practical indicator for expressing the distribution type.

The evolution of a single vein can be divided into three stages: propagation, wear and dilation. Each stage is characterized by the development of one or several types of brecciation. Mineralization marks these stages and displays different textures as a function of pressure variations. The transition from one stage to another is marked by either a mechanical discontinuity or a hydraulic continuity threshold.

The use of quantitative parameters that can distinguish between different kinds of hydrothermal breccias may lead to a better understanding of the physical processes acting in the vein environment, including structural setting, crustal level and fluid–rock interactions.

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Appendix A

This appendix presents methods for computing the geometric parameters used in this paper. Griffiths (1952) stated that all the physical properties of a sedimentary rock can be thought of as functions of the shape, size, orientation, mineral composition and packing of grains. A breccia can be considered to be a special kind of sedimentary rock, and since we limit our analysis to monolithic breccias, only four parameters are needed in order to fully describe the breccia geometry: (1) fragment morphology; (2) particle size distribution; (3) fabric; and (4) dilation ratio. Morphology is relevant to individual particles whereas size distribution is applied to the entire fragment population. Fabric defines the orientation of the fragments in the breccia and their spatial organization. Dilation ratio is the ratio between void and fragment volumes. Voids may eventually be filled by hydrothermal minerals, and its volume can be used to distinguish between matrix-supported and clast-supported breccias. Analysis generally is limited to two dimensions, using image analysis of cemented breccia slabs. Analyses of breccia textures require preliminary image processing to correct for defects or to enhance some aspect of the image, as well as for recognition of the fragments within the matrix. Ultimately the image is reduced to the features of interest only, although it may still require further editing (for instance to separate touching fragments).

Because most geological studies of breccias are done at a scale that varies between several μm (microscopic scale) and tens of meters (outcrop or mine-scale), and because of their inherent self-similarity (Sammis and Biegel, 1986), it is appropriate to choose scale independent geometric parameters. Fractal geometry is a very effective way to characterize objects with large-scale variations and it has already been used to describe particles in a fragmented media (Turcotte, 1986; Kaye, 1989). Two fractal dimensions will be used: D_r is the morphology of the fragments (roughness) and D_s is the particle size distribution (PSD; Blenkinsop, 1991).

A.1. Morphology

Particle morphology has been the object of numerous quantitative studies. For instance, Orford and Whalley (1987) have reviewed techniques that could be used to define particle shapes in sedimentology, but the names for the various shape factors are not universally consistent. Sphericity or aspect ratio (length/width) is one of the most common, allowing isometric and anisometric fragments to be distinguished (Budkewitsch and Robin, 1994; Genna et al., 1996), although it gives little information about the overall shape. Circularity (ratio between area and perimeter) gives an indication of the general shape, but not the morphological details (Burkhard, 1990; Russ, 1995). Fourier analysis has been successfully applied to sedimentology (Clarke, 1981), but is difficult to apply to highly irregular grains because the unrolling technique may induce error and gives a complex array of parameters.

The boundary fractal dimension is a relatively easy parameter to measure and it evaluates the complexity of the outline. Several techniques may be used for this type of calculation. The Mandelbrot (1975) method consists of evaluating the length of the fragment's perimeter by 'walking' along the perimeter using 'strides' of different lengths. A log-log plot of perimeter versus stride length allows the fractal dimension to be calculated. This method, however, is sensitive to artifacts (Orford and Whalley, 1987; Power et al., 1988) and a most robust approach is the Euclidean distance mapping method (Russ, 1995; Bérubé and Jébrak, 1996; Bérubé and Jébrak, submitted). Ribbons of increasing thickness

are computed from the particle outline. The area of each ribbon is then plotted against its thickness and displayed on a log–log plot. A straight line is indicative of a fractal geometry. D_r is computed using $D_r = 2 - r$, where r is the slope of the plot (Kaye, 1989). The more complex the boundary, the higher the fractal dimension. Fig. 2 gives an example of this type of calculation.

A.2. Distribution

The particle-size distribution of a breccia can also be analyzed by several methods used in sedimentology and volcanology (Sammis and Osborne, 1982). Size can be defined by several parameters, including the volume or the surface area of the fragments, and their mass or their equivalent volume diameter (Scarlett, 1991). In image analysis, the easiest parameter to determine is the cross-sectional surface, which is calculated using the total number of pixels. The particle size distribution (PSD) in brecciated rocks can be fitted by different distribution functions: exponential distribution (Brown et al., 1983), logarithmic normal distribution (Epstein, 1947), and power law (Hartmann, 1969), all of which can be computed using cumulative or noncumulative methods (Korcak, 1938; Sammis and Biegel, 1989). Two endmembers of the distribution can be recognized: Poissonian and fractal. Poissonian distribution is characterized by fragments with a characteristic mean and random variations about the mean. Fractal distribution is scale-independent and does not have any characteristic mean because it varies with the method of determination and the scale of observation. It has been demonstrated that fractal distribution is equivalent to the classic laws of fragmentation, like the Rosin–Rammler distribution and the Weibull law for example (Harris, 1966; Blenkinsop, 1991), which are based on measurements of fragment size during metallurgical ore processing. Fractal dimension of the PSD (D_s) therefore appears to be the best parameter for providing a unique value for any given breccia and for directly reflecting the degree of uniformity of the fragment size (approximately equivalent to the sorting notion). D_s is defined in the same way as D_r , but using a cumulative distribution curve of the fragment sizes (Fig. 4). A straight line will be indicative a power law distribution. Low D_s values (weak

slopes of the cumulative curve) indicate an almost isometric distribution whereas high values denotes a large variation in the size and the abundance of the fragments. This parameter is therefore a very good way to approximate PSD, although it does have some limitations: (1) particle size may not always be easy to define, especially in two dimensions (Allen, 1968), (2) the reduction of a particle-size distribution to only one parameter is obviously an approximation.

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